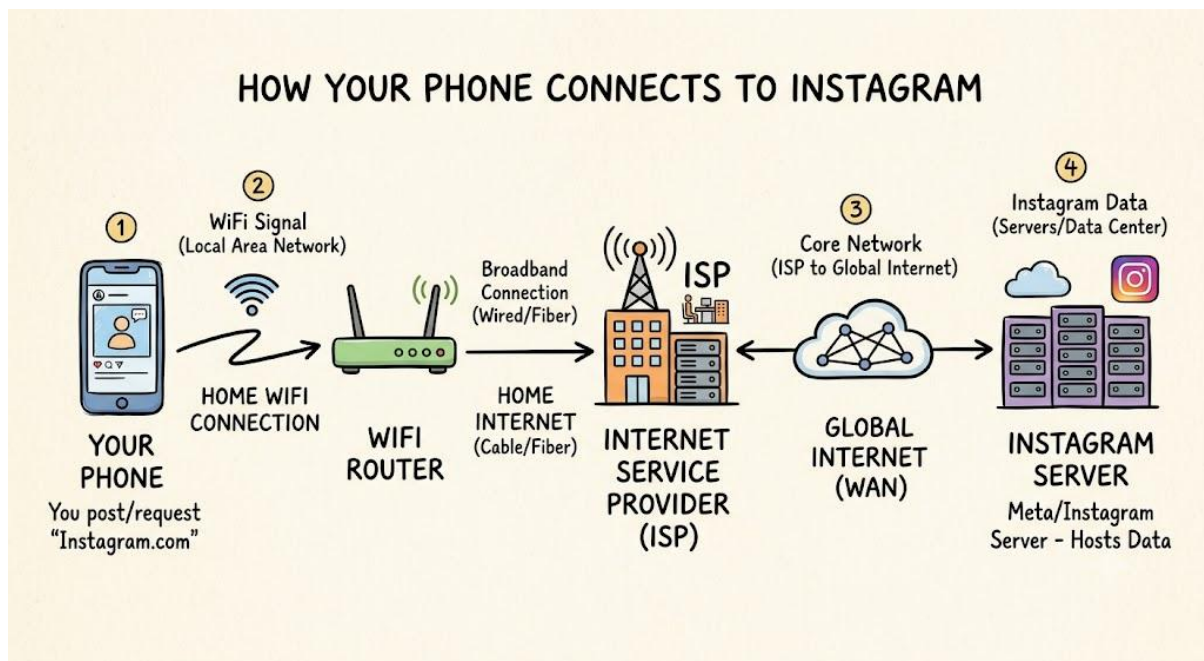


TASK-1



TASK-2

Here are 5 everyday devices and the types of networks they connect to:

1. Smartphone

Network Type: **PAN** (Personal Area Network), **LAN** (Local Area Network), and **WAN** (Wide Area Network)

How it connects:

Connects to accessories like wireless earbuds or a smartwatch via Bluetooth (**PAN**).

Connects to home/office WiFi (**LAN**).

Connects to cellular networks (4G/5G) to access the internet anywhere (**WAN**).

2. Smartwatch

Network Type: **PAN** (Personal Area Network) / **LAN**

How it connects: Primarily pairs directly with your smartphone via Bluetooth to sync data and notifications over short range (**PAN**). Many models can also connect directly to your home WiFi network (**LAN**).

3. Laptop Network Type: LAN (Local Area Network)**How it connects:** Connects via WiFi or an Ethernet cable to your local home, office, or school router, sharing resources (like printers or local storage) with other nearby devices.

4. Smart TV

Network Type: LAN (Local Area Network)

How it connects: Connects to your local home network via WiFi or Ethernet to stream media from local media servers or request content through your internet gateway.

5. Home WiFi Router

Network Type: WAN (Wide Area Network) & LAN

How it connects: Acts as a bridge between networks. It manages your internal local network (**LAN**) for all household devices and connects externally to your ISP's network (**WAN**) via fiber or coaxial cables to access the global internet.

TASK-3

1. The Internet (Public Access)

The Internet is a global, publicly accessible network of connected computers. Anyone with a data connection or WiFi password can access its public resources.

College Example: When you connect to your college's student WiFi and open **Google, YouTube, or Wikipedia**, you are using the Internet.

Access: Open to everyone worldwide.

Purpose: Sharing global information, public web browsing, and external communication.

2. An Intranet (Private Access)

An intranet is a private, restricted network that uses the same technologies as the Internet (like websites and IP addresses), but it is isolated so **only authorized users inside the organization** can access it.

College Example: Your college's internal **Student Portal, Internal Exam Results System, or Campus Library Server**.

Even if someone outside the campus knows the web address for the internal faculty portal, they cannot open it from their home internet unless they connect through a secure network tunnel (like a VPN). The server is locked strictly within the college's private network.

Access: Restricted to specific users (e.g., students, staff, faculty). **Purpose:** Securely sharing internal data, attendance records, exam papers, and administrative files without exposing them to the entire world.

TASK-4

Here is how data travels when you search for food in **Zomato**:

1. **Initiating the Request:** When you type "Pizza" and tap 'Search' in Zomato, your phone packages this request along with your location coordinates into a digital data packet.
2. **Local Transmission:** The packet travels via radio waves (**PAN/LAN**) from your phone to your home WiFi router or nearby cellular cell tower.
3. **ISP Routing:** Your router or mobile carrier (**ISP**) routes this request through physical fiber-optic cables across the internet backbone (**WAN**).
4. **Server Processing:** The request reaches Zomato's cloud servers, which search their database for nearby pizza places, prices, and ratings.
5. **The Response:** Zomato's server sends the structured results back through the exact same path to your phone, where the app renders the list of restaurants on your screen.